

Automatic differentiation

Automatic differentiation

The idea is to just write a function,
.... and its derivative is automatically
calculated!

To do this, we just need to keep track
of the variable and its derivative for
EVERY operation

```
struct Dual <: Number
    v::Float64 #value of the function
    ∂::Float64 #its derivative
end
```

x

(value, derivative)

Suppose we want to differentiate

$$f(x) = 3x + \sin(x^2)$$

```
struct Dual <: Number
    v::Float64 #value of the function
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end
```

x

(value, derivative)

```
struct Dual <: Number
    v::Float64 #value of the function
    ∂::Float64 #its derivative
end
```

```
x=Dual(1,1)
```

x

(value, derivative)

x^2

$\left(x = 1, \frac{\partial x}{\partial x} = 1\right)^2$

$\left(1^2, 2 \times 1^{(2-1)} \times 1\right)$

$(1, 2)$

```
struct Dual <: Number
    v::Float64 #value of the function
    ∂::Float64 #its derivative
end
```

```
x=Dual(1,1)
```

```
^(a::Dual,n::Integer)=Dual(a.v^n,n*a.v^(n-1)*a.∂)
```

$\sin(x^2)$

$\sin(1, 2)$

$(\sin(1), \cos(1) \times 2)$

$(0.841\ldots, 1.08\ldots)$

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struct Dual <: Number
    v::Float64 #value of the function
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end
```

```
x=Dual(1,1)
```

```
^(a::Dual, n::Integer)=Dual(a.v^n, n*a.v^(n-1)*a.∂)
```

```
sin(a::Dual)=Dual(sin(a.v), cos(a.v)*a.∂)
```

$\sin(x^2)$
(0.841..., 1.08...)

```
struct Dual <: Number
    v::Float64 #value of the function
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sin(a::Dual)=Dual(sin(a.v),cos(a.v)*a.∂)
```

Suppose we want to differentiate

$$f(x) = 3x + \sin(x^2)$$

x

$$\left(x = 1, \frac{\partial x}{\partial x} = 1\right)$$

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struct Dual <: Number
    v::Float64 #value of the function
    ∂::Float64 #its derivative
end
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```
x=Dual(1,1)
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```
^(a::Dual,n::Integer)=Dual(a.v^n,n*a.v^(n-1)*a.∂)
```

```
sin(a::Dual)=Dual(sin(a.v),cos(a.v)*a.∂)
```

3x

$$3 \times \left(x = 1, \frac{\partial x}{\partial x} = 1 \right)$$
$$\left(3x, 3 \frac{\partial x}{\partial x} \right)$$
$$(3, 3)$$

```
struct Dual <: Number
    v::Float64 #value of the function
    d::Float64 #its derivative
end
```

```
x=Dual(1,1)
```

```
^(a::Dual,n::Integer)=Dual(a.v^n,n*a.v^(n-1)*a.d)
```

```
sin(a::Dual)=Dual(sin(a.v),cos(a.v)*a.d)
```

```
*(a::Number,b::Dual)=Dual(a*b.v,a*b.d)
```

Suppose we want to differentiate

$$f(x) = 3x + \sin(x^2)$$

$$3x + \sin(x^2)$$

(3,3)

(0.841..., 1.08...)

(3 + 0.841..., 3 + 1.08...)

(3.841..., 4.08...)

```
struct Dual <: Number
    v::Float64 #value of the function
    ∂::Float64 #its derivative
end
```

```
x=Dual(1,1)
```

```
^(a::Dual,n::Integer)=Dual(a.v^n,n*a.v^(n-1)*a.∂)
```

```
sin(a::Dual)=Dual(sin(a.v),cos(a.v)*a.∂)
```

```
*(a::Number,b::Dual)=Dual(a*b.v,a*b.∂)
```

```
+(a::Dual,b::Dual)=Dual(a.v+b.v,a.∂+b.∂)
```


Let's see how you can use Forward Mode automatic differentiation to calculate gradients of a function with more than one variable

$$g(x, y) = x^2 + y^3$$

To calculate $\frac{\partial g}{\partial x}(x = 1, y = 2)$

```
g(x,y)=x^2+y^3
```

```
x=Dual(1.0,1.0)
```

```
y=Dual(2.0,0.0)
```

```
g(x,y)
```

```
g(x,y) | Dual(9.0, 2.0)
```

$$\left(x = 1, \frac{\partial x}{\partial x} = 1 \right)$$

$$\left(y = 2, \frac{\partial y}{\partial x} = 0 \right)$$

Note that $\frac{\partial y}{\partial x} = 0$ because y is Independent of x

To calculate $\frac{\partial g}{\partial x}(x = 1, y = 2)$

```
g(x,y)=x^2+y^3
```

```
x=Dual(1.0,1.0)
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```
y=Dual(2.0,0.0)
```

```
g(x,y)
```

$$\left(x = 1, \frac{\partial x}{\partial x} = 1 \right)$$

$$\left(y = 2, \frac{\partial y}{\partial x} = 0 \right)$$

← Note that $\frac{\partial y}{\partial x} = 0$ because y is
Independent of x

```
g(x,y) | Dual(9.0, 2.0)
```

$$(1,1)^2 + (2,0)^3$$

```
^(a::Dual,n::Integer)=Dual(a.v^n,n*a.v^(n-1)*a.d)
```

$$(1,2) + (8,0) = (9,0)$$

To calculate $\frac{\partial g}{\partial y}(x = 1, y = 2)$

```
g(x,y)=x^2+y^3
```

```
x=Dual(1.0,0.0)
```

```
y=Dual(2.0,1.0)
```

```
g(x,y)
```

```
g(x,y) | Dual(9.0, 12.0)
```

$$\left(x = 1, \frac{\partial x}{\partial y} = 0 \right)$$

$$\left(y = 2, \frac{\partial y}{\partial y} = 1 \right)$$

← Note that $\frac{\partial x}{\partial y} = 0$ because y is Independent of x

$$(1,0)^2 + (2,1)^3$$

$$^{(a :: \text{Dual}, n :: \text{Integer})} = \text{Dual}(a.v^n, n * a.v^{(n-1)} * a.d)$$

$$(1,0) + (8,12) = (9,12)$$